

The Machinery of Calculus

STUDENT EDITION · with Professor Logic

Before you take calculus: see the machine underneath it. Who built it, why it broke, how it got fixed — and why you'll never have to memorize a rule you can derive.

PROFESSOR LOGIC

Welcome back — or welcome, if you're new. Quick introductions: I teach machines. Not metal ones. *Reasoning* machines. And today's machine is the most famous one ever built: **calculus**.

Here's what nobody tells you before calc class. The word "calculus" is just Latin for *pebble* — the little stones people counted with. A calculus is literally **a system for calculating**. Nothing mystical. And like every calculating system, it's built from exactly three choices — the same three knobs from the logic book. If you know the knobs, calculus stops being a wall of rules to memorize and becomes a machine you can *operate*. And take apart. And audit.

The three knobs, one more time

THE CONTROL PANEL (FROM THE MATH OF LOGIC)

V — the value space: what values are allowed to exist in your system.

G — the operations: the machinery that combines values into answers.

θ (theta) — the threshold: the line that decides what *counts*.

In the logic book, θ decided which truth-values counted as "yes." In calculus, θ becomes something even better: **a tolerance — how close is close enough**. Hold that thought. It's the key to the whole subject.

PROFESSOR LOGIC · RIGOR CHECK

Every important claim in this book wears a trust label, same as before: **FORCED** means we derive it and you can check every step; **STIPULATED** means somebody chose it and it could have been otherwise; **HISTORY** means it actually happened; **ANALOGY** means it's a bridge to walk across, not a house to live in. And the standing rule: **a calculus is a map, not the territory**. Even this one. *Especially* the famous one.

THE JOURNEY

- 1 Slope: The Machine That Works — what your current tools already do perfectly
- 2 The Problem That Breaks It — a curve, one point, and division by zero
- 3 The First Machine (and Its Leak) — Newton, Leibniz, and the ghost values
- 4 The Repair: θ as Tolerance — limits, and the game you can always win
- 5 The Derivative: G on a New Value Space — deriving rules instead of memorizing them
- 6 The Reverse Gear: Accumulation — and the day the two gears meshed
- 7 Strengths, Weaknesses, and the Vindicated Ghost — where it shines, where it honestly fails, Robinson's revenge
- 8 Your Final Challenge — build, derive, audit

CHAPTER 1

Slope: The Machine That Works

Before we break anything, let's admire a machine that runs perfectly.

PROFESSOR LOGIC

Calculus is about **change** — how fast things change, and how change adds up. So let's start with the simplest change-measuring machine there is: *slope*. You've met it. Rise over run. How much does the output climb when the input steps forward?

Let's set it up as a proper machine, knobs and all, because that habit is about to save us.

THE CHALKBOARD — The slope machine **STIPULATED**

V: the number line — every ordinary number, positive, negative, decimal

G: $\text{slope}(\text{two points}) = \text{rise} / \text{run} = (y_2 - y_1) / (x_2 - x_1)$

θ: exactness — an answer counts when the arithmetic lands on it

Example: the line $y = 3x + 1$, using points (1, 4) and (5, 16):

$$\text{slope} = (16 - 4) / (5 - 1) = 12 / 4 = 3$$

Any two points on the line give the same answer. Try it. That sameness is WHY "the slope of a line" is a single number.

TL;DR

Slope = **how much up per step across**. For a straight line the machine is complete: pick any two points, divide, done. One number describes the steepness everywhere. This machine has no problems. Enjoy that. It won't last.

PROFESSOR LOGIC

Real-world reading, because slope is everywhere once you see it: your download at a steady 5 MB per second — slope 5. A car cruising at 60 km/h — slope 60 on the distance-vs-time graph. Your allowance growing by the same amount weekly — constant slope. **Straight line = constant rate = one number tells the whole story.**

TRY IT YOURSELF — CHAPTER 1

Only uses: rise over run on straight lines. Nothing else exists yet.

1. Find the slope of the line through (2, 7) and (6, 19).
2. A game character gains XP on a perfectly straight graph: (0 min, 0 XP) and (10 min, 250 XP). What's the slope, and what does it *mean* in game-speak?
3. Check the "sameness" claim: for $y = 2x + 3$, compute the slope using points (0,3) and (1,5), then again using (10,23) and (50,103). Same answer?
4. Your friend computes a slope of 0. What does the line look like? What if the run — the bottom of the fraction — were 0 instead? (Don't solve it. Just say what the machine tries to do.)

ANSWERS

1. $(19-7)/(6-2) = 12/4 = 3$.
2. $250/10 = 25$ XP per minute. That's your farming rate.
3. $(5-3)/(1-0) = 2$ and $(103-23)/(50-10) = 80/40 = 2$. Same. Any two points on a line agree — that's what makes it a line.
4. Slope 0 = flat line — the output never climbs. Run 0 = the machine tries to *divide by zero*, and division by zero has no value in V. The machine jams. Remember this jam. It's Chapter 2's whole plot.

CHAPTER 2

The Problem That Breaks It

A curve, a single point, and the most productive jam in the history of mathematics.

PROFESSOR LOGIC

Now the world gets realistic. Almost nothing changes at a constant rate. A skateboarder rolling down a ramp *speeds up*. Your phone battery drains fast at first, then slower. A ball you throw rises, hangs, falls. Their graphs aren't lines. They're **curves**. And on a curve, the steepness is *different at every point*.

So here's the question that launched the whole subject: **how steep is a curve at exactly one point?** Not on average between two points — *right there*, at that instant. The speedometer question: not "you averaged 40 km/h this trip" but "you are doing 40 *right now*."

THE CHALKBOARD — Feed the curve to the slope machine **FORCED** verified: `calc_2_1`

Curve: $y = x^2$. Question: how steep at the point $x = 3$?

The machine demands TWO points. Fine — take $x = 3$ and a second point a little ahead, at $x = 3 + h$ (h is the step size):

$$\begin{aligned}\text{slope} &= ((3+h)^2 - 3^2) / ((3+h) - 3) \\ &= (9 + 6h + h^2 - 9) / h \\ &= (6h + h^2) / h \\ &= 6 + h \quad \dots \text{as long as } h \text{ is NOT zero.}\end{aligned}$$

$h = 1$ $\rightarrow$ slope 7 (too far ahead: overshoots)
 $h = 0.1$ $\rightarrow$ slope 6.1 (closer...)
 $h = 0.001$ $\rightarrow$ slope 6.001 (closer still...)
 $h = 0$ $\rightarrow$ slope = 0/0. **JAM.** The one point we actually
 wanted is the one point the machine cannot do.

TL;DR

To get the steepness *at* a point, you want the two points to be the *same* point — and then rise over run becomes 0/0, which isn't a value in V. The answers clearly **crowd toward 6...** but the machine can't say "6." It jams exactly at the finish line.

PROFESSOR LOGIC · RIGOR CHECK

Stop and appreciate what kind of problem this is. It is not a *mystery*. It's an **engineering jam**: our G (rise over run) demands two distinct points, and our question needs one. The value space V has no value for $0/0$ — and that's not V being defective; $0/0$ genuinely has no consistent value (if $0/0 = k$, then $0 = 0 \cdot k$ works for *every* k , so no single answer is forced). The question "how steep right here" is meaningful. Our current machine just can't compute it. In the logic book, when a machine couldn't do a job, we didn't panic — we **re-engineered a knob**. Same move coming.

The stakes: why the 1600s needed this machine HISTORY

1600s. This wasn't a puzzle for fun. Kepler had shown planets move in *ellipses*, speeding up near the sun — curved motion, changing rate. Galileo had shown falling objects accelerate — again, changing rate. Navigation, artillery, astronomy, optics: the era's biggest problems all reduced to "how fast is this changing *right now*?" and "how much does this changing thing *add up to*?" Whoever built the machine that answered those would hold the keys to physics. Two people built it at almost the same moment — and hated each other for the rest of their lives over it.

TRY IT YOURSELF — CHAPTER 2

Only uses: the slope machine on two points, algebra you just watched, and the step size h .

1. For $y = x^2$ at $x = 5$, compute the two-point slope with $h = 1$, then $h = 0.1$, then $h = 0.01$. What number are the answers crowding toward?
2. Do the algebra like the chalkboard did: simplify $((5+h)^2 - 25)/h$. (Expand, cancel, divide by h .) Your answer should have an h in it.
3. Set $h = 0$ in your simplified answer from question 2. Now explain the cheat you just committed: at which earlier step did you *use* the fact that h wasn't 0?

ANSWERS

1. $h=1$: $(36-25)/1 = 11$. $h=0.1$: $(26.01-25)/0.1 = 10.1$. $h=0.01$: 10.01 . Crowding toward **10**.

2. $(25 + 10h + h^2 - 25)/h = (10h + h^2)/h = \mathbf{10 + h}$.

3. $10 + 0 = 10$. The cheat: dividing by h (going from $(10h+h^2)/h$ to $10+h$) is only legal when $h \neq 0$. So you treated h as *not zero* to divide, then as *zero* to finish. Both. In one calculation. Congratulations — you've just reproduced, by yourself, the exact move that made calculus work for 150 years *and* the exact move that made it scandalous. Next chapter: the people who did it first, and the man who called them out.

CHAPTER 3

The First Machine (and Its Leak)

Two rivals, one invention, and a value space that didn't stock the values the machine ran on.

Engineer #1: Isaac Newton, and the plague years HISTORY

1665-66, England. Cambridge University shut down — the plague was killing thousands — and a 23-year-old student went home to his family farm at Woolsthorpe with nothing to do. In about eighteen months of isolation, **Isaac Newton** worked out the core of what he called the method of *fluxions*: a machine for computing how "flowing quantities" change instant by instant. He built it because his physics demanded it — you cannot do gravity, orbits, or acceleration without it. Then, being Newton, he mostly *didn't tell anyone* for decades. He was secretive, prickly, and sat on the greatest mathematical invention of the age like a dragon on gold.

Engineer #2: Gottfried Leibniz, and the notation that won HISTORY

1675-76, Germany/Paris. A decade later, the philosopher-diplomat-mathematician **Gottfried Wilhelm Leibniz** built the same machine independently — and, unlike Newton, designed it to be *used by other people*. He invented the notation you'll actually see in class: dy/dx for the slope machine and the long-S $\int$ for accumulation. Leibniz published in **1684**; Newton's fury at being "beaten to print" ignited the nastiest priority war in the history of science — accusations of plagiarism, rigged committees, national feuds between English and continental mathematics that lasted a century. Modern verdict: they invented it independently. Newton first in private, Leibniz first in public — and Leibniz's notation was better engineering, which is why you'll write dy/dx and not Newton's dots.

PROFESSOR LOGIC

So how did their machine handle the jam — the 0/0 at the finish line? With a bold, strange move. They said: let h be **infinitely small**. Not zero — you can still divide by it. But so small that when you're done dividing, you can *throw it away* like it's nothing. Leibniz called these ghosts *infinitesimals*. Watch the machine run:

THE CHALKBOARD — The infinitesimal machine, running `FORCED` verified: `calc_3_1`

Slope of $y = x^2$ at ANY point x , the Newton–Leibniz way:

step 1:	$((x+h)^2 - x^2)/h$	(h "infinitely small")
step 2:	$= (2xh + h^2)/h$	
step 3:	$= 2x + h$	← divided by h: $h \neq 0$ here
step 4:	$= 2x$	← discarded h: $h = 0$ here

ANSWER: the slope of x^2 at any x is $2x$. At $x=3$: slope 6. ✓
Matches everything Chapter 2's numbers crowded toward. The machine WORKS. It answers every question physics asks. And...

THE CHALKBOARD — ...the machine leaks `FORCED` verified: `calc_3_2`

AUDIT: what kind of value is h ?

To divide by it (step 3): $h \neq 0$. So h is some positive amount.
To discard it (step 4): $h = 0$. So h is nothing at all.
 V is the number line. Is there a number that is
BOTH a positive amount AND nothing at all?
NO. No such value exists in V . The machine runs on a value
its own value space does not stock.

Same shape as the logic book's product-AND trap: reads beautifully, and the audit finds a value outside V .

TL;DR

Newton and Leibniz got the *right answers* with a machine that used a ghost value: h had to be "not zero" in one step and "zero" in the next. Their value space — ordinary numbers — contains no such citizen. The machine works. The machine leaks. Both true at once.

The auditor: Bishop Berkeley and the ghosts HISTORY

1734. The philosopher **George Berkeley** published a pamphlet called *The Analyst* that did to calculus exactly what our closure audit did to the product-AND machine. He asked: what *are* these quantities that are nonzero when you divide and zero when you discard? His phrase is one of the great burns in intellectual history: they are, he wrote, "*the ghosts of departed quantities.*" And here's the part that matters: **he was right**. The mathematicians of the day could not answer him. They kept using the machine anyway — for 150 years — because it kept producing correct physics. Bridges got built on it. Planets got predicted with it. A leaky machine, running the world.

PROFESSOR LOGIC • RIGOR CHECK

This is the most important engineering lesson in this whole book, so let me be precise about what it is and isn't. **"It gets right answers" and "it passes the audit" are different properties.** The infinitesimal machine had the first without the second. That's not a reason to sneer at Newton and Leibniz — building the working machine first and finding the clean foundation later is how *most* great engineering happens. But it IS a reason never to confuse the two properties. When someone shows you a system that "just works," the audit question is still owed: *what values is it secretly running on, and does its V actually stock them?* For calculus, answering that question took a century and a half. Next chapter: the fix — and it's a θ move.

TRY IT YOURSELF — CHAPTER 3

Only uses: the two-point slope algebra from Chapter 2, run the Newton-Leibniz way.

1. Run the infinitesimal machine on $y = x^2$ at $x = 7$: expand $((7+h)^2 - 49)/h$, divide by h , then discard h . What slope do you get? Check it against $2x$.
2. Run it on $y = x^3$ at a general x . You'll need $(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$. Simplify the quotient, divide by h , discard every term that still has an h . What's the slope machine's answer for x^3 ?
3. Now be Berkeley. Write one sentence, in your own words, naming exactly which two steps of your question-2 calculation contradict each other about what h is.

ANSWERS

1. $(49+14h+h^2-49)/h = (14h+h^2)/h = 14+h \rightarrow \text{discard} \rightarrow \mathbf{14}$. And $2x$ at $x=7$ is 14. ✓
2. $((x^3+3x^2h+3xh^2+h^3) - x^3)/h = (3x^2h+3xh^2+h^3)/h = 3x^2 + 3xh + h^2 \rightarrow \text{discard } h\text{-terms} \rightarrow \mathbf{3x^2}$.
3. Something like: "Dividing by h required $h \neq 0$, but discarding $3xh + h^2$ required $h = 0$ — the calculation demands both, and no number in V is both." If you wrote that, you independently reproduced *The Analyst*. It took the world 150 years to answer you. It'll take us one chapter.

CHAPTER 4

The Repair: θ as Tolerance

The fix that saved calculus wasn't a new value. It was a new threshold — and you already know how to play it.

PROFESSOR LOGIC

Here's the beautiful thing about the repair: it doesn't add ghost values to V . The number line stays exactly as it was. Instead, it changes what an *answer* is — it re-engineers θ , the threshold.

In the logic book, θ decided which values count as "yes." The repair says: for questions like ours, an answer counts when the outputs **stay within any tolerance you demand**. Not "reaches the value at $h = 0$ " — the machine still jams there, forever, and we stop pretending otherwise. Instead: "gets within tolerance θ of the value, and *stays* within it, for every small enough h ." That is the whole idea called a **limit**. Let me show you it as a game, because it genuinely is one.

THE CHALKBOARD — The tolerance game STIPULATED — the repaired θ

Claim: "as h shrinks, $6 + h$ settles at 6." (our x^2 at $x=3$ machine)

THE GAME — you doubt me, so you set the tolerance:

YOU: "get within $\theta = 0.1$ of 6."

ME: "done — for every h smaller than 0.1, $6+h$ is within 0.1 of 6."

YOU: " $\theta = 0.001$."

ME: "done — every h smaller than 0.001."

YOU: " $\theta =$ one billionth."

ME: "every h smaller than one billionth. Next."

I can meet EVERY tolerance you will ever name. That — nothing more, nothing mystical — is what "the limit is 6" MEANS:

for any $\theta > 0$ you demand, small-enough h keeps $6+h$ within θ of 6.
 h is never zero. Nothing is infinitely small. No ghosts.

TL;DR

A **limit** is a promise you can check: *name any tolerance, and the outputs get within it and stay within it*. The answer "6" doesn't come from plugging in $h = 0$ (illegal, always was). It comes from winning the tolerance game against every possible challenger. θ went from "exact hit" to "beats every tolerance" — and that one knob-turn is the entire repair.

PROFESSOR LOGIC

An analogy — **ANALOGY**, so walk across it and wave goodbye. Think of a wireless speaker as your friend walks it toward the router. There may be interference at the exact spot where the router sits — maybe the speaker can never occupy *that spot*. Doesn't matter. If for any signal strength you demand, there's a distance inside which the connection is always at least that strong, the pairing is perfect for every purpose. The limit is the signal the approach guarantees — not the value at the unreachable spot.

THE CHALKBOARD — The repaired machine, full run **FORCED** verified: calc_4_1

Slope of $y = x^2$ at x , cleanly:

1. two-point slope, $h \neq 0$ ALWAYS: $((x+h)^2 - x^2)/h = 2x + h$
2. tolerance game on $2x + h$:
distance from $2x$ is $|2x + h - 2x| = |h|$
you demand $\theta \rightarrow$ every $|h| < \theta$ wins. Every θ beatable.
3. verdict: the limit is $2x$.

Same answer Newton got. ZERO illegal steps. h was never 0; nothing "infinitely small" was ever invoked. V is untouched — the repair lives entirely in θ .

Berkeley's audit, re-run on this machine: **no leaks. CLOSED.**

The repair crew: Cauchy and Weierstrass **HISTORY**

1821-1870s. The tolerance idea was forged by **Augustin-Louis Cauchy** in Paris (his 1821 textbook started the cleanup) and finished with full precision by **Karl Weierstrass** in Berlin — a man who spent years as a small-town schoolteacher, doing world-changing mathematics at night, before anyone noticed him. In class you'll one day meet their tolerance written as the Greek letter ε ("epsilon"). When you do, smile: **ε is our θ** . The famous, feared "epsilon-delta definition" is exactly the tolerance game you just played — you name ε , I name how close h must be. You already know it. You learned it as a game, which is what it is.

PROFESSOR LOGIC · RIGOR CHECK

Notice what the repair did and did not claim, because this is anti-reification in action. It did *not* claim h "becomes" zero, or "reaches" anything, or that infinity was tamed. It **replaced a claim about an unreachable value with a claim about beatable tolerances** — something checkable, finite step by finite step. And notice the price tag, because everything costs: the repaired machine gives answers only where the tolerance game is *winnable*. Where no single value beats every tolerance, the machine now says "no limit" — honestly — instead of hallucinating a ghost. We meet such a place in Chapter 7, and the machine's honesty there is a feature, not a failure.

TRY IT YOURSELF — CHAPTER 4

Only uses: two-point slope algebra and the tolerance game as just defined.

1. Play a round: I claim $10 + h$ settles at 10 (that's Chapter 2's x^2 at $x=5$). You demand $\theta = 0.05$. What h -range meets your demand? Could you have named a θ I couldn't beat?
2. Repair your own Chapter 3 result: for $y = x^3$, the two-point slope was $3x^2 + 3xh + h^2$. At $x = 1$ this is $3 + 3h + h^2$. Play the game with $\theta = 0.5$: show that $h = 0.1$ gets within tolerance of 3. (Compute $3 + 0.3 + 0.01$ and check its distance from 3.)
3. In one sentence: what exactly changed between Newton's machine and the repaired one — V , G , or θ ? And what didn't change at all?

ANSWERS

1. Every h with $|h| < 0.05$ keeps $10+h$ within 0.05 of 10. And no — for any $\theta > 0$ you name, $|h| < \theta$ wins. The game is unloseable for me, which is precisely what "the limit is 10" asserts.
2. $3 + 0.3 + 0.01 = 3.31$; distance from 3 is $0.31 < 0.5$. Within tolerance. (Smaller θ just needs smaller h — the h -terms shrink to beat any demand.)
3. θ changed: "exact value at $h=0$ " became "beats every tolerance as h shrinks." V didn't change (same number line, no ghosts added) and G 's algebra didn't change (same expand-and-divide, now with $h \neq 0$ stated honestly). One knob. The whole repair.

CHAPTER 5

The Derivative: G on a New Value Space

The machine gets a name, a bigger V — and you stop memorizing forever.

PROFESSOR LOGIC

Look at what we built. Feed in the curve x^2 , out comes the slope-recipe $2x$ — a slope for *every* point at once. Feed in x^3 , out comes $3x^2$. Do you see what the machine actually is now? It doesn't eat numbers and produce numbers. It eats a **whole function** and produces a **whole function**.

That's a knob change worth announcing. The value space V has been upgraded: it's not the number line anymore — **V is now the space of functions themselves**.

And our G is an operator on that space: it maps each curve to its slope-recipe. This machine has a name: the **derivative**. The slope-recipe of f is written f' ("f prime"). That's the entire secret of the scary word. Derivative = the function-eating slope machine.

THE CHALKBOARD — The derivative, as three knobs STIPULATED — the upgraded machine

V: functions (curves) — x^2 , x^3 , $5x^2 + x$, all of them
G: derivative: $f \rightarrow f'$, where $f'(x)$ = the tolerance-game verdict of $(f(x+h) - f(x))/h$ as h shrinks
θ: the tolerance game from Chapter 4 — unchanged, doing all the work

So far, derived by hand (not memorized — DERIVED):

$$x^2 \rightarrow 2x \qquad x^3 \rightarrow 3x^2$$

PROFESSOR LOGIC

Now for the payoff I promised on the cover: **no memorization**. In calc class you'll be shown "the power rule": the derivative of x^n is $n \cdot x^{n-1}$. Most students memorize it like a magic spell. You're going to *watch it fall out of the algebra*, because you've already done it twice. The whole trick is one fact about expanding $(x+h)^n$.

THE CHALKBOARD — The power rule, derived not decreed **FORCED** verified:
calc_5_1

Expand $(x+h)^n$ and watch the pattern:

$$\begin{aligned}(x+h)^1 &= x + 1 \cdot h \\(x+h)^2 &= x^2 + 2x \cdot h + h^2 \\(x+h)^3 &= x^3 + 3x^2 \cdot h + 3xh^2 + h^3 \\(x+h)^4 &= x^4 + 4x^3 \cdot h + 6x^2h^2 + 4xh^3 + h^4\end{aligned}$$

Every row: first term x^n , second term $n \cdot x^{n-1} \cdot h$, then terms with h^2 , h^3 ... Now run the slope machine on x^n :

$$\begin{aligned}\frac{(f(x+h) - f(x))}{h} &= \frac{(x^n + n \cdot x^{n-1} \cdot h + h^2(\text{stuff}) - x^n)}{h} \\&= n \cdot x^{n-1} + h \cdot (\text{stuff})\end{aligned}$$

The x^n cancels. Dividing by h strips one h off everything left. The h -terms lose the tolerance game (they shrink below any θ). Sole survivor: $n \cdot x^{n-1}$. That's the "power rule" — not a spell, a CANCELLATION PATTERN you can rerun any time you doubt it.

TL;DR

Derivative of $x^n = n \cdot x^{n-1}$, because in $(x+h)^n$ only the $n \cdot x^{n-1} \cdot h$ term survives the cancel-divide-shrink gauntlet. $x^5 \rightarrow 5x^4$. $x^{10} \rightarrow 10x^9$. You never memorized it; you watched every other term die. **A rule you can derive is a receipt. A rule you memorize is a debt.**

PROFESSOR LOGIC

Two more receipts and you can differentiate every polynomial on Earth. First: the derivative of a **sum** is the sum of the derivatives — because $(f+g)$'s slope quotient literally splits into f 's quotient plus g 's quotient; no step forces them to interact. Second: constants ride along — the slope quotient of $5 \cdot f$ is 5 times the quotient of f , so $5x^2 \rightarrow 5 \cdot (2x) = 10x$. These aren't extra rules to store. They're what the fraction does when you write it down.

PROFESSOR LOGIC · RIGOR CHECK

Honesty about scope, because a receipt says exactly what was bought: we derived the power rule for *counting-number* powers — $n = 1, 2, 3, \dots$ — since our expansion pattern is only established there. In calc class the rule gets extended to negative and fractional powers, and it holds there too, but *we haven't derived that*, so we don't claim it. **STIPULATED** until you earn it. Feel the difference? Most books hand you the biggest version of a rule and hope. We hand you the version we paid for, and the tools to buy the rest.

TRY IT YOURSELF — CHAPTER 5

Only uses: the expansion pattern, cancel-divide-shrink, and the sum/constant receipts above.

1. Derive the derivative of x^4 yourself using the $(x+h)^4$ row from the chalkboard: subtract x^4 , divide by h , run the tolerance game. What survives?
2. Using your receipts (power rule for $n = 1, 2, 3, 4$, sums, constants): find f' for $f(x) = x^3 + x^2$, then for $f(x) = 4x^3$, then for $f(x) = 2x^4 + 7x$.
3. The receipt test: a classmate says "the derivative of x^6 is $6x^5$, because the teacher said so." You say "it's $6x^5$ because—" and finish the sentence with the *mechanism*, in one line, no authority allowed.

ANSWERS

1. $(4x^3h + 6x^2h^2 + 4xh^3 + h^4)/h = 4x^3 + 6x^2h + 4xh^2 + h^3$. Every h -term loses the tolerance game. Survivor: $4x^3$.
2. $x^3+x^2 \rightarrow 3x^2 + 2x$. $4x^3 \rightarrow 4 \cdot 3x^2 = 12x^2$. $2x^4+7x \rightarrow 2 \cdot 4x^3 + 7 \cdot 1 = 8x^3 + 7$.
3. "—because in $(x+h)^6$ the only term with exactly one h is $6x^5 \cdot h$, and after subtracting x^6 and dividing by h , everything else still carries an h and shrinks below any tolerance." Authority not required. That's the whole point of a receipt.

CHAPTER 6

The Reverse Gear: Accumulation

The other half of calculus — and the day someone noticed the two machines were one machine.

PROFESSOR LOGIC

Flip the question. The derivative asks: given the curve, how fast is it changing? The reverse asks: **given the rate, how much piles up?** Your speedometer readout for a whole trip — how far did you go? Water flowing into a tank at a changing rate — how full is it? Rate in, total out. This is **accumulation**, and in class it'll be called the *integral*, written with Leibniz's long S: $\int$. (It's an S for "sum." That's all the symbol means. A stretched-out letter S.)

And here's the good news: you already own the machinery. Accumulation is thin slices plus the tolerance game — the same θ doing the same job.

THE CHALKBOARD — Accumulation by slices **FORCED** verified: calc_6_1

Speed $f(t) = 2t$ (going faster and faster). Distance from $t=0$ to $t=x$?

Slice time thin. In each slice, pretend speed is constant (it barely changes in a thin slice), so slice-distance $\approx$ speed $\times$ width. Add the slices. Thinner slices $\rightarrow$ smaller pretending $\rightarrow$ and the totals win the TOLERANCE GAME against a single number.

For $f(t) = 2t$ there's a shortcut we can fully trust: the region under the line from 0 to x is a TRIANGLE — base x , height $2x$.

$$\text{accumulated total} = \frac{1}{2} \cdot \text{base} \cdot \text{height} = \frac{1}{2} \cdot x \cdot 2x = x^2$$

Now the thunderclap. We accumulated the rate $2t$ and got x^2 . Chapter 3 derived: the slope machine on x^2 gives back... $2x$. Accumulate the rate $\rightarrow x^2$. Differentiate $x^2 \rightarrow$ the rate. The machines UNDO each other.

TL;DR

Accumulation = slice thin, multiply, add, tolerance-game the total. And the punchline of the entire subject: **the slope machine and the accumulation machine are reverse gears of each other**. Speed is the slope of distance; distance is the accumulation of speed. In class this is the *Fundamental Theorem of Calculus* — the discovery that made two hard problems into one machine running forwards and backwards.

The prehistory: Archimedes plays the tolerance game — 1,900 years early

HISTORY

~250 BC, Syracuse. Archimedes wanted the area of a circle — a curved region no formula of his day could touch. His move: trap it. Draw polygons *inside* the circle (too small) and *outside* it (too big), then add sides — 6, 12, 24, 96 — squeezing the truth between bounds that close in from both sides. Name any tolerance; enough sides beats it. He called it the *method of exhaustion*, and it is our θ -game, played two millennia before ε was hired. When Newton and Leibniz built accumulation by slices, they were mechanizing Archimedes' squeeze. When Cauchy and Weierstrass repaired the foundations, they were, in a sense, writing down what Archimedes had been careful about all along.

PROFESSOR LOGIC · RIGOR CHECK

What we actually verified versus what we waved at — say it plainly. We *verified* the gear-meshing for one case, $f(t) = 2t$, using honest triangle geometry: accumulate $\rightarrow x^2$, differentiate $\rightarrow 2t$. ✓ The claim that this meshing holds for *all* the curves you'll meet is the Fundamental Theorem, and its full proof needs machinery we haven't built — so here it's **STIPULATED**, with one case paid in full as evidence. Your calc class will pay the rest. Keep the habit: one verified case plus a named IOU beats a confident wave every time.

TRY IT YOURSELF — CHAPTER 6

Only uses: rectangle and triangle areas, plus the derivative receipts from Chapter 5.

1. Constant speed $f(t) = 3$ from $t = 0$ to $t = x$. The region under the graph is a rectangle. Accumulate it (area formula), then differentiate your answer. Do the gears mesh — do you get 3 back?
2. Speed $f(t) = t$ (half the ramp-up of the chalkboard). The region from 0 to x is a triangle with base x , height x . Accumulate; differentiate; check the mesh.
3. Your friend says integrals are "a totally separate topic from derivatives — twice the memorizing." Using your two mesh-checks, tell them the actual relationship in one sentence.

ANSWERS

1. Rectangle: $3 \cdot x = 3x$. Derivative of $3x$ is 3. ✓ Meshed.
2. Triangle: $\frac{1}{2} \cdot x \cdot x = \frac{1}{2}x^2$. Derivative: $\frac{1}{2} \cdot 2x = x$. ✓ Meshed.
3. "They're one machine in two gears — accumulation undoes differentiation and vice versa — so every derivative receipt you own is already an integral receipt read backwards." Half the memorizing your friend feared, and (with your receipts) approximately none.

CHAPTER 7

Strengths, Weaknesses, and the Vindicated Ghost

Where the machine shines, where it honestly says "no" — and the 20th-century plot twist Berkeley never saw coming.

PROFESSOR LOGIC

Time to grade the machine like engineers grade machines: what's it *for*, where does it fail, and what did every feature cost?

Strengths first, because they're staggering. The repaired calculus is arguably the most successful tool humans have ever built. Anywhere something changes *smoothly* — orbits, currents, epidemics, waves, markets, rockets, the cooling of your coffee — this machine turns "how fast?" and "how much?" into computable answers. Physics as we know it is written in it. Your phone's signal processing runs on its descendants. One machine, three knobs, four centuries of civilization built on top.

THE CHALKBOARD — Where the machine honestly refuses **FORCED** verified: `calc_7_1`

The corner. $f(x) = |x|$ (the absolute-value V-shape). Slope at $x = 0$?

Play the tolerance game from BOTH sides of 0:

from the right ($h > 0$): $(|0+h| - 0)/h = h/h = +1$

from the left ($h < 0$): $(|0+h| - 0)/h = -h/h = -1$

For a limit to exist, ONE value must beat every tolerance for ALL small h . Demand $\theta = 0.5$: no single number is within 0.5 of both +1 and -1 (they're 2 apart). **Unwinnable. NO limit.**

The machine's verdict at the corner: "no slope here."

Not an error. Not a mystery. The correct answer, honestly given — a corner genuinely HAS no single steepness.

TL;DR

The derivative machine works on **smooth** curves. At a sharp corner it says "no answer" — and that's the machine being *truthful*, because a corner points two directions at once. A machine that refuses impossible questions is a better machine than one that hallucinates. (Remember the repaired θ 's price tag from Chapter 4? This is it, being paid — visibly, honestly.)

PROFESSOR LOGIC

And it gets wilder, filed under **HISTORY** because we cite it rather than build it: Weierstrass — the same repairman — constructed a curve that is *all corners*: connected everywhere, yet so infinitely jagged that the slope machine returns "no answer" at **every single point**. Mathematicians called it a monster. It's also a warning label: "smooth enough for calculus" is a real requirement, not a formality. Most textbook curves qualify. Reality sometimes doesn't — coastlines, stock charts, static — and knowing the machine's operating range is part of owning the machine.

The plot twist: Abraham Robinson vindicates the ghosts **HISTORY**

1960s. Here's the ending nobody expected. Three centuries after Berkeley's audit, the logician **Abraham Robinson** — using the machinery of modern mathematical logic — built a *bigger value space*: the **hyperreal numbers**. In that V , infinitesimals are legal citizens: genuine values smaller than every ordinary positive number yet not zero, with fully consistent rules for computing — including a lawful version of "discard the h at the end." Inside Robinson's carrier, Newton and Leibniz's original style of calculation is **completely rigorous**. It's called *non-standard analysis*, and it produces the same answers as the limit machine. The ghosts weren't nonsense after all. They were citizens of a value space nobody had built yet.

PROFESSOR LOGIC · THE ANTI-REIFICATION FINALE

Sit with what Robinson's result means, because it's this book's deepest lesson wearing a calculus costume. We now have **two different machines** for calculus: Weierstrass's (ordinary V , repaired θ — the tolerance game) and Robinson's (expanded V with infinitesimals, simpler θ). Same answers. Same physics. So — which one is what calculus *really is*? **Wrong question.** That question assumes there's one true machine hiding behind the mathematics, and there isn't — there are engineered carriers, with different V 's and θ 's, that both do the job. " dx " is not a mystical object you must believe in; it's machinery whose meaning depends on which carrier you're running. Your class runs Weierstrass's, and now you know it's a *choice* with a history, a price, and an alternative. The map is not the territory. Not even this map. Not even the famous one.

TRY IT YOURSELF — CHAPTER 7

Only uses: the tolerance game, both-sides checking, and the receipts you own.

1. Run the both-sides check on $f(x) = |x|$ at $x = 3$ (not at the corner): compute $(|3+h| - 3)/h$ for a small positive h (say 0.1) and a small negative h (say -0.1). Do the two sides agree now? What's the slope there?
2. A classmate claims: "the slope of $|x|$ at 0 is 0, because the point at the bottom is flat." Audit this with the tolerance game: what do the two sides actually report, and why does no single value win?
3. Grading time. Fill in one line each: the machine's greatest strength; its honest limitation; and the sentence you'd say to someone who insists "the limit definition is what calculus truly is."

ANSWERS

1. $h = 0.1$: $(3.1 - 3)/0.1 = 1$. $h = -0.1$: $(2.9 - 3)/(-0.1) = 1$. Both sides say 1. Away from the corner, $|x|$ is just the line $y = x$, slope 1. The machine works fine everywhere the curve is smooth.
2. Right side reports +1, left side -1 . For any $\theta < 1$, no value is within θ of both. The bottom point isn't "flat" — it's a disagreement between two directions, and the machine correctly refuses to average a disagreement into a fake answer.
3. Strength: turns smooth change into exact computation — the engine of modern science. Limitation: refuses (honestly) wherever smoothness fails — corners, jumps, monsters. And: "The limit definition is one excellent carrier for calculus — Robinson built another with legal infinitesimals that gets the same answers. Carriers are engineered, not revealed."

CHAPTER 8

Your Final Challenge

Everything you need, nothing you don't. Build it yourself.

PROFESSOR LOGIC

Look back down the road. You started with rise-over-run. You watched it jam on a curve. You ran Newton's leaky ghost-machine and got right answers anyway. You *were* Berkeley and audited it. You repaired it by re-engineering θ into a tolerance game. You upgraded V to functions and derived — not memorized — the power rule. You meshed the two gears. You found the corner where the machine honestly refuses, and met the bigger carrier where the ghosts turned out to be legal.

That's not "pre-calc." That's the **machinery** of calc. When class hands you dy/dx in the fall, you'll be the one who knows what's under the hood, whose notation it is, why it survived its audit — and which rules you can rebuild from scratch when memory fails. Which is all of them.

THE WHOLE BOOK, IN ONE BREATH

A calculus is a machine: V , G , θ . The first one leaked. The repair moved θ from "exact hit" to "beats every tolerance." Every rule is a receipt you can re-derive. And no carrier — not even the famous one — is the territory.

YOUR FINAL CHALLENGE — FULL MACHINE, FROM SCRATCH

Uses everything in the book. Nothing new. Show every step.

1. **Derive.** From the two-point slope, derive the derivative of $f(x) = x^2 + 3x$ completely: expand, cancel, divide ($h \neq 0$, stated out loud), then win the tolerance game. No receipts allowed — raw machinery only. Then check your answer *against* your receipts.
2. **Play θ .** Your derivation left you with something of the form (answer) + h . An examiner demands tolerance $\theta = 0.003$ at $x = 1$. Name the h -range that wins, and explain in one sentence why every possible θ is beatable.
3. **Audit.** A textbook from 1710 says: "Let h be an infinitely small quantity; then $2x + h = 2x$ exactly." Write the two-line Berkeley audit of this sentence, then the one-line Weierstrass repair of it, then the one-line Robinson rehabilitation of it.
4. **Mesh the gears.** Accumulate the constant rate $f(t) = 5$ from 0 to x , differentiate the result, and confirm the mesh. Then state which single case of the Fundamental Theorem you have now personally verified, and label the general theorem with the honest tier.
5. **Name the price.** One sentence: what did the θ -repair cost the machine? (Chapter 7 showed you the receipt.)
6. **Stay honest.** One sentence: why is "which carrier is the TRUE calculus?" the wrong question — and what's the right one?

WHAT A STRONG ANSWER LOOKS LIKE

1. $((x+h)^2 + 3(x+h) - x^2 - 3x)/h = (2xh + h^2 + 3h)/h = 2x + 3 + h$ (dividing legal since $h \neq 0$). Tolerance game: $|2x+3+h - (2x+3)| = |h|$ beats any θ once $|h| < \theta$. Verdict: $f'(x) = 2x + 3$. Receipts check: $x^2 \rightarrow 2x$, $3x \rightarrow 3$, sum $\rightarrow 2x + 3$. ✓
2. Every $|h| < 0.003$ wins. Any θ is beatable because the error IS $|h|$, and h can always be taken smaller than any positive number named — that unloseable game is what "limit" means.
3. Audit: "You divided by h , so $h \neq 0$; you then set $2x+h = 2x$, so $h = 0$ — no value in V is both." Repair: "Replace 'equals exactly' with 'stays within any demanded tolerance of $2x$ for small enough $h \neq 0$.'" Rehabilitation: "Or move to Robinson's hyperreal V , where a nonzero infinitesimal h with lawful discard-rules is a legal citizen — then the 1710 sentence becomes rigorous as written."
4. Rectangle: $5x$. Derivative: 5 . ✓ Meshed. Verified case: constant rates (and, from Ch 6, the linear rate $2t$). General FTC: **STIPULATED** here — one paid case, the rest owed, and I know exactly who'll collect (my calc teacher).
5. The repaired machine only answers where the tolerance game is winnable — at corners and jumps it now says "no answer," trading total coverage for total honesty.
6. Because carriers are engineered tools, not discovered truths — two different (V , θ) setups compute identical physics. The right question is the engineer's: *which carrier fits this job?*

PROFESSOR LOGIC

That's the machine. When it's September and someone next to you groans "I'll never memorize all these rules," you can slide over and say the most subversive sentence in mathematics: "*you don't have to — I'll show you where they come from.*"

Go be that person. And when any system — mathematical or otherwise — asks you to accept its ghosts without an audit, you know the drill.

Run the audit. Every time. Even on the famous ones.

A note for teachers and the curious. Chalkboards stamped with test names (like `calc_5_1`) refer to the companion verification code of the Math of Logic project, where each stamped computation is checked symbolically and numerically. The history is standard: Newton's fluxions in the plague years 1665–66 (published much later); Leibniz's independent development 1675–76, notation dy/dx and $\int$, published 1684; the priority dispute; Berkeley's *The Analyst* (1734) and "the ghosts of departed quantities"; Cauchy's *Cours d'analyse* (1821); Weierstrass's ϵ - δ formulation and his continuous nowhere-differentiable function; Archimedes' method of exhaustion; Abraham Robinson's non-standard analysis (1960s), which makes infinitesimal methods rigorous in the hyperreal field. The mathematics on every FORCED board is complete-derivation checkable by hand — which is the point. As Professor Logic insists: check it yourself.